MA 26200: Linear Algebra and Differential Equations

The information in this syllabus complements the information in the course-wide syllabus at `https://www.math.purdue.edu/academic/courses/semester/202620/ma26200/`. If information in this syllabus contradicts the course-wide syllabus, then the information in the course-wide syllabus should be regarded as the correct one.

Instructor Information

- Name: Dr. Gert Vercleyen
- Office: MATH 409
- Email: gvercley@purdue.edu
- Office Hours: Tuesday and Thursday 4:30 PM - 6:00 PM. You can also email me in advance to arrange a meeting if none of these times work. It is not possible to arrange recurring meetings.

Course Information

- MA 26200 - Linear Algebra and Differential Equations
- Credit Hours: 4.00. Linear algebra, elements of differential equations. Not open to students with credit in MA 26500 or 26600. 4.000 Credit hours.
- Prerequisites: Undergraduate level MA 26100 Minimum Grade of C- or Undergraduate level MA 18200 Minimum Grade of C- or Undergraduate level MA 17400 Minimum Grade of C- or Undergraduate level MA 27100 Minimum Grade of C- or Undergraduate level MA 27101 Minimum Grade of C- or Undergraduate level MATH 26100 Minimum Grade of C- or Undergraduate level MA 17200 Minimum Grade of C- or Undergraduate level MA 26300 Minimum Grade of C minus
- **Time:** 1:30 PM-2:45 PM
- **Classroom:** ME 2061

- Course Syllabus and Ground Rules: see `https://www.math.purdue.edu/academic/courses/semester/202620/ma26200/`

- Course Calendar: available on Brightspace (Content → calendar)

Learning Resources

- Required text – Edwards/Penney/Calvis: Differential Equations & Linear Algebra, 4th Edition.
- MyLab is required. Students will use it to do online homework.
- An electronic version of the textbook comes with MyLab, and a hard copy of the textbook is not required.
- Students should access MyLab though their recitation Brightspace page. Their lecture page is not linked with MyLab.
- We recommend that students should buy a MyLab access code when they access it from their recitation Brightspace page. That way they know they are buying the correct code.
- Students get a two-week grace period at the beginning of the semester. After the grace period expires, they have to buy access to MyLab.
- MyLab access is based on the textbook. Students who bought MyLab for Calculus will have to buy it for MA262, even if their Calculus book code has not expired.
- The MA262 textbook is not used in any other math class at Purdue. Students only need to buy access for one semester.
- Students can find qualified tutors through the Mathematics Department Home page. But keep in mind that your professors do not select those tutors. If you want to hire a tutor, feel free to ask your professor to recommend someone from that list.
- Current MA262 Teaching Assistants cannot be hired as paid private MA262 tutors even for students who are not in their sections. This would a violation of university regulations.

Course Description

This course aims to provide students with a foundation in the theory of finite-dimensional linear algebra and ordinary differential equations.

Learning Outcomes

- Learn the basic properties of matrices.
- Learn how to solve systems of linear equations.
- Learn the basic properties of determinants.
- Learn about linear independence, spanning sets and bases in the context of vector spaces.
- Learn the statement of the theorem of existence and uniqueness of solutions of first order differential equations and standard methods of solving linear, and some special non-linear, differential equations of first order.
- Learn the standard methods of solving linear differential equations of constant coefficients of arbitrary order.
- Learn basic numerical methods for solving differential equations.
- Learn about the applications of differential equations to mechanical, physical, and biological models.
- Learn the theory of eigenvalues and eigenvectors and how it applies to systems of differential equations.

Assignments

The due dates for assignments are available on Mylab.

- There will be 36 online homework assignments. Students will do the homework using MyLab.
- TAs will assign online homework and due dates will be stated in MyLab.
- The lowest three homework grades will be dropped.
- Partially done homework will be graded accordingly.
- Students will get a score of zero on homework they did not attempt to do.
- Students who want to appeal a homework score due to possible computer error should contact their TA.
- **Late homework**: Except in cases of University approved reasons, late homework will not be accepted.
- **Collaboration on assignments**: Collaboration and discussion with peers on both online and written homework are not only permitted but also encouraged. However, the work submitted should be a genuine representation of your understanding. Mindless copying of homework is considered as academic dishonesty and will hinder your learning progress.

- **Use of AI and web material**: The usage of AI such as chatgpt and websites such as wolframalpha are permitted as long as no exercises are used as litteral input to these websites. The exercises are copyrighted material and should not be shared on the internet. Note that mindlessly copying results from websites is also considered academic dishonesty and hinders your learning progress. Moreover, it is important to note that while AI chatbots might be good at presenting results with confidence, they are terrible at mathematical reasoning.

Quizzes

- TAs will prepare and give weekly quizzes (except during the first week, exam weeks and quiet period) during recitation according to what has been covered in lecture and recitation.
- Students who are entitled to special accommodations will get the appropriate time and conditions for quizzes. They need to speak with their TA to make arrangements.
- TAs will also manually grade the quizzes.
- The lowest three quiz grades will be dropped.

Exams

- There will be two multiple-choice midterm exams and one multiple-choice final exam all in person.
- Exams will be computer graded.
- Students who are entitled to special accommodations will get the appropriate time and conditions for exams from the DRC. See more details in the course-wide syllabus.
- Exam 1: Feb 19, 8:00 P.M to 9:00 P.M. PHYS 114, UC114, Covers Lessons 1 to 14.
- Exam 2: Apr 14, 8:00-9:00 P.M. PHYS 114, PHYS 112. Covers lessons 15 to 26.
- Final Exam (date TBA by Purdue). The final exam covers all lessons.
- No exam grades will be dropped.

Grading scale and score distribution

Assignments	Due	Points
Online homework	throughout semester	Average=100
Quizzes	during recitation throughout semester	Average=100
Two (one-hour long) midterm exams	Dates and lessons: see above	100 each
One (two-hour long) Final exam	during finals week	200
		Total: 600

Grading scale

- A+: 570 – 600
- A: 540 – 569
- A-: 510 – 539
- B+: 480 – 509
- B: 450 – 479
- B-: 420 – 449
- C+: 390 – 419
- C: 360 – 389
- C-: 330 – 359
- D+: 300 – 329
- D: 270 – 299
- No D minus in MA262
- F: 269 or below
- Borderline cases (this means within one point of a cut-off, say for example 419.5) will be considered on a case-by-case basis

These cut-offs may be lowered at the end of the semester, but they cannot be raised. For example, a student with 538 points will get at least A minus, but it is possible they actually get A, if these cut-offs are adjusted. Historically these cut-offs have not been adjusted by more than a few points.

Accommodations for Students with Disabilities

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone at 765-494-1247. If you have been certified by the Disability Resource Center (DRC) as eligible for accommodations, you should contact your instructor to discuss your accommodations as soon as possible. Here are instructions for sending your Course Accessibility Letter to your instructor: `https://www.purdue.edu/drc/students/course-accessibility-letter.php`

Disclaimer

This syllabus is subject to change. Any changes will be announced in Brightspace.